ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** acidic	Acid turns blue litmus red and red remains red.
2	**(d)** NaOH	NaOH.
3	**(a)** it contains acetic acid	Acid.
4	**(a)** basic	Typical of bases.
5	**(a)** sodium sulphate	$\mathrm{Na_2SO_4}$.
6	**(b)** it is a base	Base.
7	**(c)** neutral	Neutral.
8	**(d)** potassium nitrate	$\mathrm{KNO_3}$.
9	**(c)** more lactic acid forms	Acid increases.
10	**(c)** soap solution	Basic.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A base (1); Soap solution (1).	1+1
12	$\mathrm{H_2SO_4 + 2\,NaOH \longrightarrow Na_2SO_4 + 2\,H_2O}$ (1); Salt: sodium sulphate (1).	1+1
13	Salt is formed in neutralisation (1); Acid is sour and turns blue litmus red (1).	1+1
14	Vinegar is an acid and neutralises the base (1); Baking soda is also a base so it cannot neutralise it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	X: basic (1); Y: acidic (1); Reason from colours (1).	1+1+1
16	Both contain bases (1); Neutralise stomach acid (1); Rate of action differs (1).	1+1+1
17	Avoid confusion (1); Safety (1); Correct use (1).	1+1+1
	OR: Boil pieces (1); Filter extract (1); Vinegar gives red, soap gives green (1).	1+1+1
18	Too much base can upset the stomach (1); Stomach acid is needed for digestion (1); A doctor's advice is safer (1).	1+1+1
	OR: Vinegar (1); Lemon (1); Baking soda (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Test with blue and red litmus (1); Two turn blue litmus red: vinegar, lemon (1); Two turn red litmus blue: lime water, soap (1); Sugar: no change (1); Do not taste the liquids (1).	1+1+1+1+1
	OR: Fizzing (1); Gas released (1); Carbon dioxide (1); Acid + base reaction (1); Lime water turns milky with the gas (1).	1+1+1+1+1
20	Add litmus to acid: red (1); Add base drop by drop (1); Colour turns violet (1); $\mathrm{HCl + NaOH \longrightarrow NaCl + H_2O}$ (1); Evaporate to get salt (1).	1+1+1+1+1
	OR: Acids: blue litmus red (1); Bases: red litmus blue (1); Neutral: no change (1); Turmeric red in base (1); China rose pink in acid (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** calcium chloride	$CaCl_2$.
2	**(c)** to neutralise mouth acids	Prevents decay.
3	**(a)** basic	Both indicate base.
4	**(a)** $HCl + KOH \longrightarrow KCl + H_2O$	Acid and base give salt and water.
5	**(a)** concentrated acid is dangerous	Safety.
6	**(c)** neutral	No colour shift.
7	**(a)** hydrogen gas	Gas is not a general product.
8	**(b)** heat produced can splash the acid	Safety rule.
9	**(b)** blue	Excess base.
10	**(a)** vinegar	A mild acid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Antacid for stomach (1); Lime for acidic soil (1).	1+1
12	It can burn skin (1); It can spoil clothes (1).	1+1
13	Acid reacts with the base (1); Carbon dioxide is released (1).	1+1
14	China rose shows base by turning green (1); Soap is basic (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Antacid is a base (1); Lime is a base (1); Both reduce acidity (1).	1+1+1
16	Neutral: no colour change (1); Acidic: blue litmus red (1); China rose pink in acid (1).	1+1+1
17	Use litmus or turmeric (1); Observe the colour change (1); Interpret (1).	1+1+1
	OR: $HCl + KOH \longrightarrow KCl + H_2O$ (1); $H_2SO_4 + 2\,KOH \longrightarrow K_2SO_4 + 2\,H_2O$ (1); Salt and water formed (1).	1+1+1
18	Dip blue and red litmus in each (1); Compare the colour changes (1); No change means neutral (1).	1+1+1
	OR: Lime water is basic (1); Vinegar is acidic (1); Different effect on indicator (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Acids: batteries, fertilisers (1); Acids: cleaning (1); Bases: soap making (1); Bases: antacids (1); Bases: whitewash (1).	1+1+1+1+1
	OR: Mix soil and water (1); Add vinegar: fizz means basic soil (1); Add baking soda: fizz means acidic soil (1); Use litmus (1); Treat accordingly (1).	1+1+1+1+1
20	Acids: red or pink (1); Neutral: purple (1); Bases: green or blue (1); Test lemon, water, soap (1); Record (1).	1+1+1+1+1
	OR: Contain acids (1); Harm fish (1); Corrode pipes (1); Add base (1); Safe release (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** colour change	Visible change.
2	**(b)** turmeric	Turmeric.
3	**(a)** slaked lime	Base.
4	**(b)** it gives salt and water	Acid + base.
5	**(d)** carbon dioxide	Fizzing.
6	**(d)** organic manure or compost	Mildly acidic manure.
7	**(c)** it loses its acidic property	Becomes neutral.
8	**(d)** vinegar and baking soda	Opposite nature.
9	**(c)** a base	Neutralisation.
10	**(d)** lemon juice, tomato juice, water	Typically.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Test both with blue litmus (1); Both turn red (1).	1+1
12	It has more water (1); Its effect on skin is weaker (1).	1+1
13	Vinegar is acidic (1); Cabbage is an indicator (1).	1+1
14	Red litmus turns blue (1); Turmeric turns red (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Neutralisation gives out heat (1); Temperature rises (1); Heat is released (1).	1+1+1
16	Acid dissolves tarnish (1); Makes surface shiny (1); Citric acid reacts (1).	1+1+1
17	Compound formed from acid and base (1); Sodium chloride (1); Also water formed (1).	1+1+1
	OR: The solution becomes basic (1); Red litmus turns blue (1); It is no longer neutral (1).	1+1+1
18	Nitric acid + potassium hydroxide gives potassium nitrate + water (1); Hydrochloric acid + calcium hydroxide gives calcium chloride + water (1); Each reaction gives a salt and water (1).	1+1+1
	OR: To neutralise acids made by bacteria (1); Protects teeth (1); Prevents decay (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Factory waste: add a base (1); Soil: slaked lime (1); Indigestion: antacid (1); Ant bite: baking soda (1); Principle: base cancels acid (1).	1+1+1+1+1
	OR: Stomach acid helps digestion (1); Too much causes acidity (1); Antacid neutralises (1); Overuse makes the stomach too basic (1); Consult a doctor (1).	1+1+1+1+1
20	Corrosive (1); Use goggles and gloves (1); Add acid slowly to water (1); Stir (1); Never add water to acid (1).	1+1+1+1+1
	OR: Taste (1); Touch (1); Litmus (1); Metals (1); Examples (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
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